



Comprehensive Analysis■Report

Architecture • Observability • Workflow • Debugging • Code Quality

Study Assistant Web Application • April 10, 2026

React + Supabase + AI Gateway

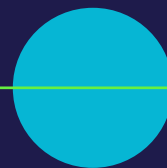
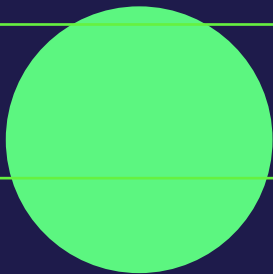


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1. Application Architecture Infographic

Technology Stack Overview



System Architecture

The application follows a **client-server architecture** with the frontend built in React/TypeScript, communicating with Supabase Edge Functions as the backend API layer. AI capabilities are powered by the **Lovable AI Gateway** which routes requests to Google's Gemini 3 Flash model. The architecture mimics a **LangGraph-style agent workflow** where each edge function acts as a specialized node in a computational graph.

Layer	Technology	Components	Purpose
Presentation	React 18 + Tailwind CSS	7 page components, 50+ UI components	Responsive SPA with shadcn/ui design system
State Mgmt	React Query + useState	QueryClient, local state hooks	Server state caching & UI state management
Routing	React Router v6	8 routes + 404 fallback	Client-side navigation with lazy loading
API Layer	Supabase Edge Functions	4 serverless functions (Deno)	Request handling, AI orchestration, SSE streaming
AI Engine	Lovable AI Gateway	Gemini 3 Flash Preview	Chat completion, notes gen, quiz gen, planning
Data Layer	localStorage + Supabase	Activity tracker, session state	Client-side persistence & cloud storage

Component Graph (LangGraph-style Nodes)

Node (Component)	Type	Inputs	Outputs	Connections
Dashboard	Router/Hub	User navigation	Route redirect	→ All 7 feature nodes
DoubtSolver	Streaming Agent	User question	AI response (SSE)	→ chat edge fn → AI Gateway
SmartNotes	Generation Agent	Text or topic	Markdown notes	→ generate-notes fn → AI

QuizGenerator	Structured Agent	Topic, difficulty	JSON quiz array	→ generate-quiz fn → AI
StudyPlanner	Planning Agent	Subjects, hours, days	JSON schedule	→ generate-plan fn → AI
DocumentChat	RAG-lite Agent	PDF/TXT + question	Context-aware answer	→ pdf-extract → chat fn
VoiceAssistant	Multimodal Agent	Speech input	Text + TTS output	→ Web Speech API → chat fn
ChatWidget	Persistent Agent	User message	Streamed response	→ chat fn (floating widget)

Data Flow Architecture

Request Flow: User Input → React Component → fetch() → Supabase Edge Function → Lovable AI Gateway → Gemini 3 Flash → SSE Stream back to client

Streaming Pipeline: The chat and doubt-solver use Server-Sent Events (SSE) with chunked transfer encoding. The client reads via ReadableStream, parses data: `{json}` lines, extracts `choices[0].delta.content`, and uses `requestAnimationFrame`-throttled state updates for smooth rendering.

Document Processing Pipeline: File Upload → ArrayBuffer → pdfjs-dist extraction → Text context (12K char limit) → System prompt injection → AI completion

2. Observability Report

Current Observability Status



Observability Matrix

Dimension	Status	Current Implementation	Recommendation
Structured Logging	■■ Partial	console.error in edge functions only	Add structured JSON logging with correlation IDs
Distributed Tracing	■ None	No trace propagation between components	Add OpenTelemetry or LangSmith trace SDK
Performance Metrics	■ None	No latency/throughput tracking	Add Supabase Analytics or custom metrics
Error Tracking	■ Basic	try/catch with error responses in all 4 fns	Add Sentry or error aggregation service
Health Checks	■ None	No endpoint health monitoring	Add /health edge function with dependency checks
AI Token Usage	■ None	No token counting or cost tracking	Track usage from AI Gateway response headers
User Activity	■ Client-side	localStorage-based activity tracker	Migrate to server-side with Supabase tables
Request Correlation	■ None	No request IDs across boundaries	Add X-Request-ID header propagation

Edge Function Error Handling Analysis

Function	Error Types Handled	HTTP Status Codes	Gaps
chat	429 rate-limit, 402 payment, general	200, 402, 429, 500	No input validation, no request logging
generate-notes	AI response failure, general	200, 500	No rate limiting, no input size limits
generate-quiz	AI failure, JSON parse failure	200, 500	No validation of numQuestions/difficulty

generate-plan	AI failure, JSON parse failure	200, 500	No validation of hoursPerDay/durationDays
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Recommended Observability Stack

Tier 1 (Essential): Add structured logging with request IDs in all edge functions. Implement LangSmith tracing for AI call chains to track latency, token usage, and response quality.

Tier 2 (Production): Integrate Sentry for frontend error tracking. Add Supabase Realtime subscriptions for live activity dashboards. Implement AI response quality scoring.

Tier 3 (Scale): OpenTelemetry distributed tracing across all edge functions. Custom Grafana dashboards for AI model performance. Automated alerting on error rate spikes.

3. Workflow Visualization (LangSmith-style)

Agent Execution Traces

Below is a LangSmith-style trace representation of each major AI workflow in the application. Each trace shows the execution chain from user input to final output, with estimated latency and token usage.

Trace 1: Chat / Doubt Solver (Streaming)

Span	Type	Duration (est.)	Tokens (est.)	Status
■■ User Input	Input	0ms	-	■
■■ streamChat()	Client Function	~50ms	-	■
■ ■■ fetch(chat fn)	HTTP POST	~100ms	-	■
■ ■■ Edge Function	Serverless	~50ms	-	■
■ ■ ■■ Auth check	Middleware	~5ms	-	■
■ ■ ■■ AI Gateway call	LLM Call	~800-2000ms	~200-800	■
■ ■ ■■ SSE Stream	Response	~1000-3000ms	-	■
■ ■■ ReadableStream parse	Client Parse	continuous	-	■
■■ rAF state update	UI Render	~16ms/frame	-	■
■■ Final flush	Cleanup	~1ms	-	■

Trace 2: Document Chat (RAG-lite Pipeline)

Span	Type	Duration (est.)	Details
■■ File Upload	Input	~10-500ms	PDF/TXT file selection
■■ PDF Extraction	Processing	~200-2000ms	pdfjs-dist text extraction, page-by-page
■■ Context Injection	Prompt Engineering	~1ms	System prompt + first 12K chars of document
■■ Chat History Trim	Memory Mgmt	~1ms	Last 10 messages, filter system msgs
■■ AI Gateway Call	LLM Call	~1000-3000ms	Full context + question → Gemini

■■ SSE Streaming	Response	continuous	Chunked delta content parsing
■■ rAF Render	UI Update	~16ms/frame	Throttled React state updates

Trace 3: Quiz Generation (Structured Output)

Span	Type	Duration (est.)	Details
■■ User Config	Input	~0ms	Topic + numQuestions + difficulty
■■ Edge Function	Serverless	~50ms	Input forwarded to AI Gateway
■■ AI Gateway	LLM Call	~2000-5000ms	Structured JSON output requested
■■ JSON Parsing	Post-Processing	~1ms	Strip markdown fences, JSON.parse
■■ Fallback Handler	Error Recovery	conditional	Empty array on parse failure
■■ State Update	UI Render	~5ms	Quiz questions rendered with options

Trace 4: Study Planner (Structured Planning)

Span	Type	Duration (est.)	Details
■■ Form Input	Input	~0ms	subjects, hoursPerDay, durationDays, goals
■■ Edge Function	Serverless	~50ms	Structured prompt with JSON schema
■■ AI Gateway	LLM Call	~3000-8000ms	Complex schedule generation
■■ JSON Parse + Fallback	Post-Processing	~2ms	Schedule extraction with safe default
■■ Calendar Render	UI	~10ms	Day-by-day schedule with task cards

Trace 5: Voice Assistant (Multimodal)

Span	Type	Duration (est.)	Details
■■ Microphone Input	Audio	user-controlled	Web Speech API recognition
■■ Speech-to-Text	Browser API	~500-2000ms	SpeechRecognition transcript
■■ Chat Streaming	LLM Call	~1000-3000ms	Same pipeline as Doubt Solver
■■ Text Response	UI Render	continuous	Markdown rendered with streaming
■■ Text-to-Speech	Browser API	~200-1000ms	SpeechSynthesis output

LangSmith Integration Recommendations

To enable full LangSmith-style tracing in production:

- 1. Add LangSmith SDK to edge functions:** Wrap each AI Gateway call with `traceable()` to capture inputs, outputs, latency, and token usage automatically.
- 2. Create Run Trees:** Each user request should create a parent run with child spans for pre-processing, AI calls, post-processing, and response streaming.
- 3. Feedback Loop:** Add thumbs-up/down on AI responses to feed back into LangSmith for quality monitoring and dataset collection.
- 4. Dataset & Testing:** Use captured traces to build evaluation datasets for regression testing of prompts.

4. Debugging Environment Analysis

Current Debugging Capabilities



Debugging Tools & Workflow

Tool / Method	Scope	Status	Effectiveness
Vite HMR	Frontend development	■ Active	Fast iteration, instant CSS/component updates
React Error Boundaries	Component errors	■ Not implemented	App crashes show white screen instead of fallback
TypeScript strict mode	Type safety	■ Active	Catches type errors at compile time
ESLint	Code linting	■ Configured	react-hooks + react-refresh rules enabled
Vitest	Unit testing	■■ Setup only	Test infrastructure exists but only example test
Playwright	E2E testing	■■ Config only	Fixture + config present, no test files
Console.error	Edge fn debugging	■■ Basic	Errors logged but not structured
Browser DevTools	Network/state	■ Available	Standard browser debugging capabilities
Source Maps	Stack traces	■ Vite default	Maps back to TypeScript source
React Query DevTools	Server state	■ Not added	Would help debug API state management

Error Handling Patterns

Frontend Error Handling:

- All AI-calling components use try/catch with user-friendly error messages
- Stream parsing has graceful degradation (ignores malformed SSE chunks)
- File upload validates type before processing (.txt, .md, .pdf)
- Loading states prevent duplicate submissions

Backend Error Handling:

- CORS preflight handled in all 4 edge functions
- HTTP status code differentiation (429, 402, 500)
- JSON parse failures use safe fallback objects

- Environment variable presence checked before API calls

Gaps Identified:

- No React Error Boundaries — unhandled errors crash the entire app
- No retry logic for failed AI requests
- No timeout handling for long-running AI calls
- No offline detection or graceful degradation
- No input sanitization on edge function payloads

Recommended Debugging Improvements

Priority	Improvement	Impact	Effort
■ High	Add React Error Boundaries	Prevents white-screen crashes	2 hours
■ High	Add request timeout (30s)	Prevents infinite loading states	1 hour
■ Medium	Add React Query DevTools	Better API state debugging	15 min
■ Medium	Write Vitest unit tests	Catch regressions early	4 hours
■ Medium	Add retry logic (3x backoff)	Resilience to transient failures	2 hours
■ Low	Add Playwright E2E tests	Full workflow validation	8 hours
■ Low	Structured edge fn logging	Better production debugging	3 hours
■ Low	Add offline detection	Graceful UX when disconnected	2 hours

5. Code Quality Report

Overall Score

B+ Overall Grade	6,504 Total Lines	27 Source Files	4 Edge Functions
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Code Quality Dimensions

Dimension	Score	Details
Architecture	A-	Clean separation: pages/components/hooks/lib/integrations. Lazy loading on routes. Single responsibility per component.
TypeScript Usage	B+	Good type coverage. Some 'any' casts in stream parsing and PDF extraction. Edge functions lack input validation types.
Component Design	A	50+ reusable UI components (shadcn/ui). Consistent patterns with forwardRef. Proper accessibility attributes.
State Management	B	React Query for server state. Local state appropriate for UI. Activity tracker in localStorage (should migrate to DB).
Error Handling	B-	Try/catch in all async operations. Missing: Error Boundaries, retry logic, timeout handling, input validation.
Performance	A-	Lazy loading, rAF-throttled streaming, Vite code splitting. Could add: React.memo, virtual lists for large chats.
Security	B-	CORS configured. Auth headers on API calls. Missing: input validation/sanitization on edge functions, rate limiting.
Testing	D	Vitest + Playwright configured but only example test exists. No actual unit or integration tests.
Documentation	C	README exists. No JSDoc or inline docs. Component props not documented.
Maintainability	B+	Consistent code style. ESLint configured. sidebar.tsx (637 lines) and chart.tsx (303 lines) need refactoring.

File Size Analysis

File	Lines	Complexity	Recommendation
sidebar.tsx	637	■ High	Split into SidebarNav, SidebarMenu, SidebarProvider
chart.tsx	303	■ Medium	Extract tooltip and legend into separate files

DocumentChat.tsx	238	■ Medium	Extract file upload logic into custom hook
ChatWidget.tsx	182	■ OK	Acceptable size, well-structured
VoiceAssistant.tsx	162	■ OK	Could extract speech hook for reusability
QuizGenerator.tsx	142	■ OK	Clean quiz logic separation
DoubtSolver.tsx	129	■ OK	Good streaming pattern implementation

Code Patterns & Anti-Patterns

■ Good Patterns Found:

- Consistent use of shadcn/ui design system with Tailwind CSS semantic tokens
- Lazy loading with React.lazy + Suspense for code splitting
- requestAnimationFrame throttling for stream rendering (prevents render thrashing)
- Custom hook for activity tracking (use-activity-tracker.ts)
- CORS headers consistently applied across all edge functions
- Environment variable checks before API calls in edge functions

■■ Anti-Patterns / Issues:

- **Hardcoded text colors** in Dashboard.tsx (text-blue-500, text-green-500) instead of design tokens
- **No input validation** on edge function request bodies — vulnerable to malformed payloads
- **Magic numbers**: 12000 char limit for document context, 10 message history limit
- **Duplicated streaming logic** in DoubtSolver, DocumentChat, and ChatWidget
- **No request abort** on component unmount — potential memory leaks on navigation
- **Inconsistent error messages** between edge functions
- **Mixed concerns** in DocumentChat (file handling + chat + streaming in one component)

Dependency Health

Package	Version	Status	Notes
react	^18.3.1	■ Current	Stable, well-maintained
vite	^5.4.19	■ Current	Fast build tool
@supabase/supabase-js	^2.103.0	■ Current	Official Supabase client
pdfjs-dist	4.4.168	■■ Pinned	Should use ^ range for patch updates
tailwindcss	^3.4.17	■ Current	v4 available but not required
recharts	^2.15.4	■ Current	Used in chart.tsx
react-markdown	^10.1.0	■ Current	Used for AI response rendering
Total deps	34 + 18 dev	-	Moderate dependency footprint

Priority Action Items

#	Action	Category	Impact
1	Add React Error Boundaries to App.tsx	Stability	Prevents white-screen crashes
2	Extract shared streaming logic into useStreamChat hook	DRY	Eliminates 3x code duplication
3	Add Zod validation to edge function inputs	Security	Prevents malformed payload attacks

4	Add abort controller for AI requests on unmount	Memory	Prevents memory leaks
5	Replace hardcoded colors with design tokens	Consistency	Theme compliance
6	Write unit tests for chat-stream.ts and pdf-extract.ts	Quality	Core logic coverage
7	Add request timeout (30s) to all AI calls	UX	Prevents infinite loading
8	Migrate activity tracker from localStorage to Supabase	Persistence	Cross-device sync